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JOHN J. FLYNN, ROBERT M. CIPOLLETTI and MICHAEL J. NOVACEK

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Chronology of early Eocene marine and terrestrial strata, Baja California, Mexico

JOHN J. FLYNN *Department of Geology, Field Museum of Natural History, Chicago, Illinois 60605-2496*
ROBERT M. CIPOLLETTI *Department of Geological Sciences, Rutgers University, New Brunswick, New Jersey 08903*
MICHAEL J. NOVACEK *Department of Vertebrate Paleontology, The American Museum of Natural History, New York, New York 10024*

ABSTRACT

We present a multidisciplinary study of the marine and terrestrial biostratigraphy, and paleomagnetism of early Eocene, intertonguing marine and continental strata from the Punta Prieta area, central Baja California, Mexico. The mammalian fauna is assigned to the Wasatchian North American Land Mammal Age, rather than the Clarkforkian assignment of earlier studies. Extensive molluscan assemblages from intertonguing marine strata are correlated to the restricted "Capay" Stage and therefore are middle Ypresian (middle early Eocene) in age. This correlation is supported by a benthic foraminiferal assemblage assigned to the Penutian/Ulatisian Stage, and the ostracode *Buntonia* (early Eocene first appearance). This study is only the second that provides direct correlation of early Paleogene marine and terrestrial biochronologies in North America. Our correlation indicates that part of the Wasatchian mammal age is correlative with the Ypresian age, as had been suggested previously from indirect evidence. All of the paleomagnetic sites from three sections are reversely magnetized; the associated middle early Eocene biochronologic information permits correlation of this reversed polarity interval to Chron C23R or the reversed polarity interval within Chron C24N. Wasatchian mammal assemblages now are known from northern Canada (Ellesmere Island) to central Baja; all available geochronologic information indicates that these assemblages are synchronous across broad, intracontinental areas. Our early Eocene paleomagnetic study provides the first early Cenozoic results from the Baja peninsula and the first direct constraint on the early Cenozoic plate-tectonic motion of the peninsula. Our results indicate that Baja California was in its pre-Gulf of California position, sutured adjacent to the northern Mexican mainland, *prior* to the middle early Eocene. These results conflict with some previous interpretations of the motion of Baja; discrepancies may be caused by failure to treat small terranes as discrete, potentially independent, mobile land masses.

INTRODUCTION

The Cenozoic geology of the Baja California peninsula initially was reported on in the late 1800s (see references in Beal, 1948) and has

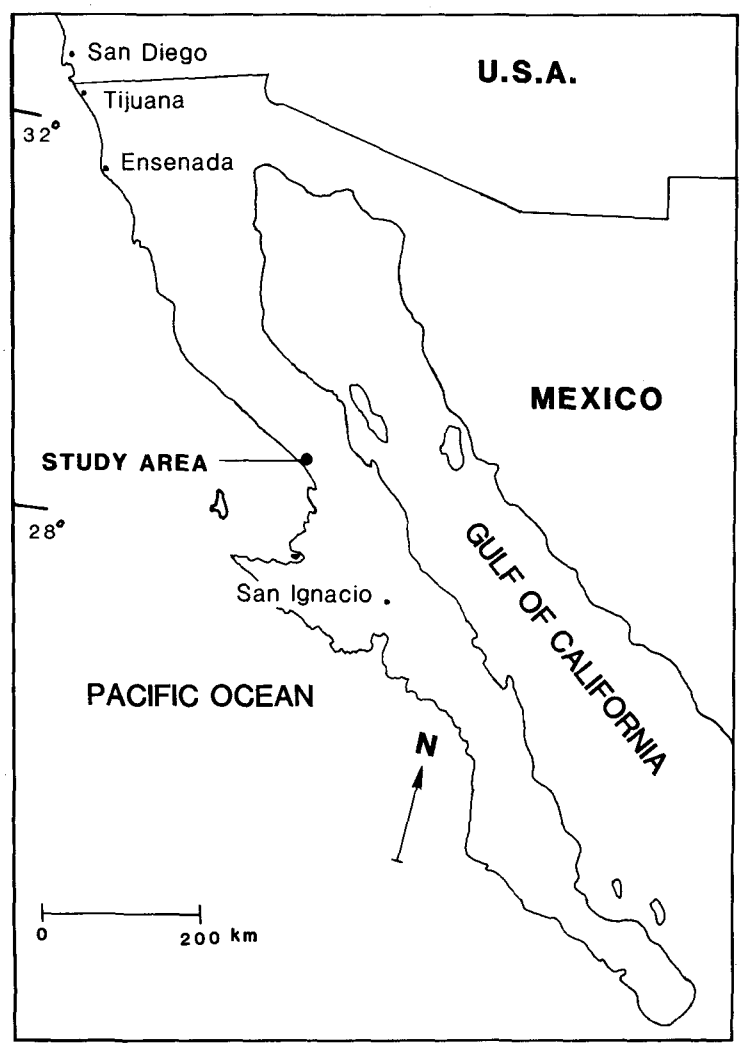


Figure 1. Map of the Baja California Peninsula, Mexico, showing the location of the study area.

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been studied more extensively since 1921 (Darton, 1921; Heim, 1922; Santillan and Barrera, 1930; Beal, 1948; Mina, 1956, 1957; Durham and Allison, 1960; Fife, 1968; Gastil and others, 1975). A number of very recent studies used paleomagnetism to constrain the plate-tectonic motion of the Baja California peninsula in an attempt to refine understanding of translation and rotation of plate-margin "terranes." Several studies have reported occurrences of terrestrial fossil mammals from Baja California (Morris, 1966; Lillegraven, 1972, 1976; Ferrusquia-Villafranca, 1978, 1984; Flynn and Novacek, 1984; Novacek and others, 1987 and in press). The early Cenozoic (Paleogene) mammal faunas are among the southernmost and westernmost occurrences of Paleogene fossil mammals in North America. Therefore, these faunas are critically important for understanding the biochronologic, paleoecologic, and paleobiogeographic history of early Cenozoic mammalian faunas. Fossiliferous terrestrial strata directly intertongue with fossiliferous marine strata in the Punta Prieta area, providing an excellent opportunity to integrate marine and terrestrial biochronologies. In this study, we present an integration of mammalian biochronology, marine microfossil and macroinvertebrate biochronology, and paleomagnetic stratigraphy for early Eocene strata from the Rosarito quadrangle, Baja California. Our data provide the first direct correlation of early Eocene marine and terrestrial biochronologies for North America; the correlation has significant implications for early Paleogene global correlation and geochronology. The paleomagnetic data also help to determine the paleogeographic history of the Baja California peninsula.

GEOLOGIC SETTING/ STRATIGRAPHY

The study area surrounds the town of El Rosario, near Bahia Santa Rosalia, within the Rosarito 7.5′ quadrangle (CETENAL, 1974, Map HIID69), Baja California Norte, Mexico (Fig. 1). The strata in the area dip gently to the west and are eroded into typical badlands topography. Local points of geographic reference include several large Mesozoic intrusive and metamorphic hills and a pair of conglomerate-capped buttes, the Lomas las Tetas de Cabra. The buttes lie ~25 km south of the town of Punta Prieta. Early Cenozoic strata rest unconformably on Mesozoic strata and crystalline basement.

We examined fossiliferous, marine, and terrestrial strata in four detailed stratigraphic sections: "Lomas las Tetas de Cabra" (LTC and GTXX, 28°41′N, 114°03′30″W), "Younger World" (YW, 28°41′N, 114°04′30″W), "SB Hill" (SB, 28°33′30″N, 114°03′W), and "Slump Hill" (SH, 28°32′30″N, 114°02′30″W), with accessory investigations in intervening areas (Fig. 2). Strata in this region previously have been referred to the Sepultura, Bateque, Tepetate, and "unnamed" formations.

The terrestrial strata are predominantly fluviatile red sandstones and siltstones with interbedded conglomerates. These units crop out in a restricted area surrounding the Lomas las Tetas de Cabra (= Occidental Buttes of Morris, 1966; Gastil and others, 1975). Novacek and others (in press) refer these strata to a new formation, the Las Tetas de Cabra Formation.

The marine strata of the study area, tentatively, and questionably, may be assigned to the Bateque Formation (of Mina, 1957; see Cipolletti, 1986). Fife (1968) correlated all Paleogene strata in the Rosarito quadrangle to the Sepultura Formation, on the basis of both lithology and inferred age. Age cannot be used in recognizing a formation, however, and sections described by Fife (1968) and this study are more similar lithologically to the type sections of the Tepetate Formation (Heim, 1922; Beal, 1948) and Bateque Formation (Mina, 1957). The type section of the Tepetate Formation is ~600 km from the study area, whereas the type section of the Bateque Formation is ~230 km distant. According to the North American Commission on Stratigraphic Nomenclature (1983), a formation should be traced physically from the type section to be valid in another area; this is not possible for any of the three formations. We questionably assign most of the marine strata in our study area to the Bateque Formation, because the lithologies of the "SB Hill" and "Slump Hill" sections are very similar lithologically to the type Bateque Formation, and the Bateque Formation type section crops out closer to the Rosarito quadrangle than that of any other formation on the peninsula.

Marine units are interpreted to have been deposited both in the near-shore marine environment and in a lagoonal (or some other protected marine embayment) type of environment (Cipolletti, 1986). These interpretations are made on the basis of analyses of lithology, sedimentology, sedimentary structures, and faunal/floral assemblages.

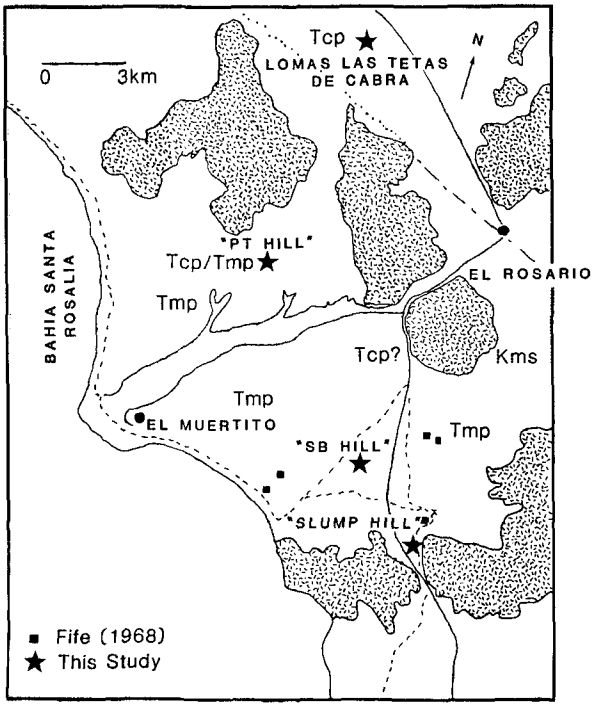


Figure 2. Geologic map of the study area, showing locations of stratigraphic sections and fossil localities (stars). Tmp, Paleogene marine units; Tcp, Paleogene terrestrial units; Kms, Cretaceous marine units.

MAMMALIAN BIOSTRATIGRAPHY

In 1966, Morris reported the first early Cenozoic mammals from Baja California, Mexico. Morris' (1966) collection from "Occidental Buttes" (Punta Prieta area) consisted of only four specimens of three or four mammalian taxa. On the basis of these taxa, Morris (1966) and Ferrusquia-Villafranca (1978, 1984) concluded that the fauna most likely was Clarkforkian (late Paleocene–early Eocene) in age; given this age determination, the fauna was believed to include the earliest perissodactyl (*Hyracotherium*). Sloan (1969) and Gingerich (1976) hypothesized that many of the mammals that first appear in the extensive Wasatchian Land

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Mammal Age (early Eocene) deposits in the United States may have evolved earlier in Mexico/Central America during the late Paleocene and migrated northward during the early Eocene. Recent work in the same area has significantly increased the mammalian fauna known from this sequence (Table 1; Flynn and Novacek, 1984; Novacek and others, 1987 and in press). This much more extensive mammalian fauna clearly is Wasatchian, rather than Clarkforkian, in age. In particular, the simultaneous first appearances of *Hyracotherium*, *Diacodexis*, and *Prolimnocyon* (hyaenodontid creodont) have been used to define the beginning of the Wasatchian (Rose, 1980, 1981; Gingerich and others, 1980), *Meniscotherium* has a Wasatchian first appearance (Rose, 1980, 1981), and *Hyopsodus* is one of the most common Wasatchian taxa (there are only two reported occurrences in Clarkforkian strata). In fact, the fauna from Lomas las Tetas de Cabra is remarkably similar to Wasatchian faunas from the Rocky Mountain basins, although there are several taxa that are endemic to this sequence. The contemporaneity and similarity of the Baja and United States faunas would negate the only evidence for previous hypotheses of a Central American origin for many of the North American Wasatchian taxa.

TABLE 1. MAMMALIAN FAUNA, LOMAS LAS TETAS DE CABRA, PUNTA PRIETA AREA

Marsupialia
Didelphidae
new genus
Condylarthra
Hyopsodontidae
cf. *Hyopsodus*
Meniscotheriidae
Meniscotherium cf. *priscum*
Phenacodontidae
Phenacodus vortmanni
Ectocion new species
Mesonychia
Mesonychidae
Dissacus sp.
?Mesonychia
Wyolestes new species
Pantodonta
Pantolambdidae
?new genus
Tillodontia
Esthonychidae
Esthonyx sp.
Carnivora
"Miacidae"
genus and species undetermined
Creodonta
Hyaenodontidae
Prolimnocyon sp.
Oxyaenidae
Oxyaena sp.
Perissodactyla
Equidae
Hyracotherium seekinsi
Brontotheriidae
Eotitanops sp.
Artiodactyla
Dichobunidae
Diacodexis sp.
Rodentia
?Paramyidae
genus and species undetermined

Note: data in this table after Novacek and others, 1987.

MARINE BIOSTRATIGRAPHY

Molluscan Faunas

Extensive molluscan assemblages were collected from the "SB Hill" and "Slump Hill" sections (Table 2; Figs. 3 and 4). Detailed comparisons of the molluscan material to published descriptions and plates, and to type specimens (originals and casts) and comparative material in the American Museum of Natural History and the Los Angeles County Museum permit identification of most of the material to species or subspecies. L. R. Saul and R. L. Squires aided in identification of questionable taxa. Age of the molluscan assemblages was determined by correlation to the standard North American Molluscan Stage zonations of Clark and Vokes (1936) and Weaver and others (1944), as modified by Givens (1974) and Squires (1984).

TABLE 2. FAUNAL LIST OF MOLLUSCA FROM THE MARINE STRATA, ROSARITO QUADRANGLE

"SB Hill" Locality	
Bivalvia	Gastropoda
Solena (Eosolen) novacularis	*Cirsotrema* sp.
Venericardia (Pacifor) durhami	*Turritella uvasana infera*
V. (P.) diabloensis	*Cerithium excelsum*
V. (P.) cf. *aragonia*	*Polinices (Euspira) nuciformis*
V. (P.) aragonia lutmani	*Calyptraea* cf. *diegoana*
V. (P.) aragonia joaquinensis	*Tympanotonos* aff. *papalis*
Corbula (Carycorbula) dickersoni	*Acteon moodyi*
Phacoides cf. *meganosensis*	*Ectinochilus* cf. *(Macilentos) macilentus*
Crassatella cf. *meganosensis*	
Crassatella sp.	
Gari eoundulata	
Ostrea cf. *angusta*	
Acanthocardia (Shedocardia) brewerii	
Macrocallista (Costacallista) domengenica capayana	
"Slump Hill" Locality	
Bivalvia	Gastropoda
Pholadomya sp.	*Turritella* cf. *andersoni*
Macrocallista ?new species	*T. meganosensis protumescens*
M. (C.) domengenica capayana	*Cerithidea* ?

The known temporal ranges of the molluscan fauna are shown in Figures 5 and 6. On the basis of the overlapping temporal ranges of both the gastropods and bivalve pelecypods, the molluscan assemblages clearly can be correlated to a restricted "Capay" Stage (of Givens, 1974, and Squires, 1984). We use the term "Capay" Stage in the restricted sense of Givens (1974); this restricted "Capay" Stage corresponds to the *Siphonalia sutterensis* zone (Clark and Vokes, 1936), and *Turritella andersoni* and *T. uvasana infera* zones (Givens, 1974). Saul (1983b) noted that *Turritella andersoni* and *Venericardia (Pacifor) hornii lutmani* indicated a restricted "Capay" Stage, as did the co-occurrence of these two species with *T. uvasana infera*, *T. meganosensis protumescens*, *T. buwaldana crooki*, and *V. (P.) aragonia joaquinensis*.

The restricted "Capay" Stage assignment is based on the presence of "Capay" Stage index taxa (restricted to the stage): *V. (P.) hornii lutmani*, *T. meganosensis protumescens*, and *T.* cf. *andersoni*, and the overlapping ranges in the Baja sections of (1) taxa with "Capay" Stage first appearances: *V. (P.) aragonia joaquinensis*, *V. (P.) aragonia*, *Acanthocardia (Schedocardia) brewerii*, *Solena (Eosolen) novacularis*, *Ectinochilus* cf. *(Macilentos) macilentus*, and *Cerithium excelsum*, and (2) taxa with "Capay" Stage last occurrences: *V. (P.) aragonia diabloensis*, *V. (P.) durhami*, *Macrocallista (Costacallista) domengenica capayana*, *T. uvasana infera*, *Crassatella meganosensis*, and *Corbula (Carycorbula) dickersoni*.

Our results indicate possible range extensions into the "Capay" Stage for several molluscan taxa. *Gari* cf. *eoundulata* and *Acteon* cf. *moodyi* previously have been identified only in the younger "Domengine" Stage. *Phacoides* cf. *meganosensis* previously was known only in the older "Meganos" Stage; however, our specimens are somewhat different than those previously assigned to *P. meganosensis*, and we feel that available information is not conclusive enough to warrant changing its known range.

Three taxa (*Tympanotomus* cf. *papalis*, *Ostrea rarilamella*, and *O.* cf. *angusta*) are very similar to, or conspecific with, taxa previously identified from the Ypresian of the Paris Basin; these occurrences suggest possible correlation with Ypresian faunas.

Of the 28 molluscan taxa identified in this study, 82% are found in other localities along the North American west coast, 10% closely resemble or are conspecific with taxa from the Ypresian of the Paris Basin, and 8% probably

represent new species. There is strong evidence, therefore, that West Coast Molluscan Stages are applicable to faunas from Baja California, at least during the early Paleogene.

Benthic Foraminifera

Six benthic foraminifera taxa have been recovered from the "SB Hill" section (Table 3). The temporal ranges of these taxa in the Paleogene strata of western North America extend from the Ynezian Stage to the Narizian Stage. *Amphistegina* has an Eocene (uppermost Yne-

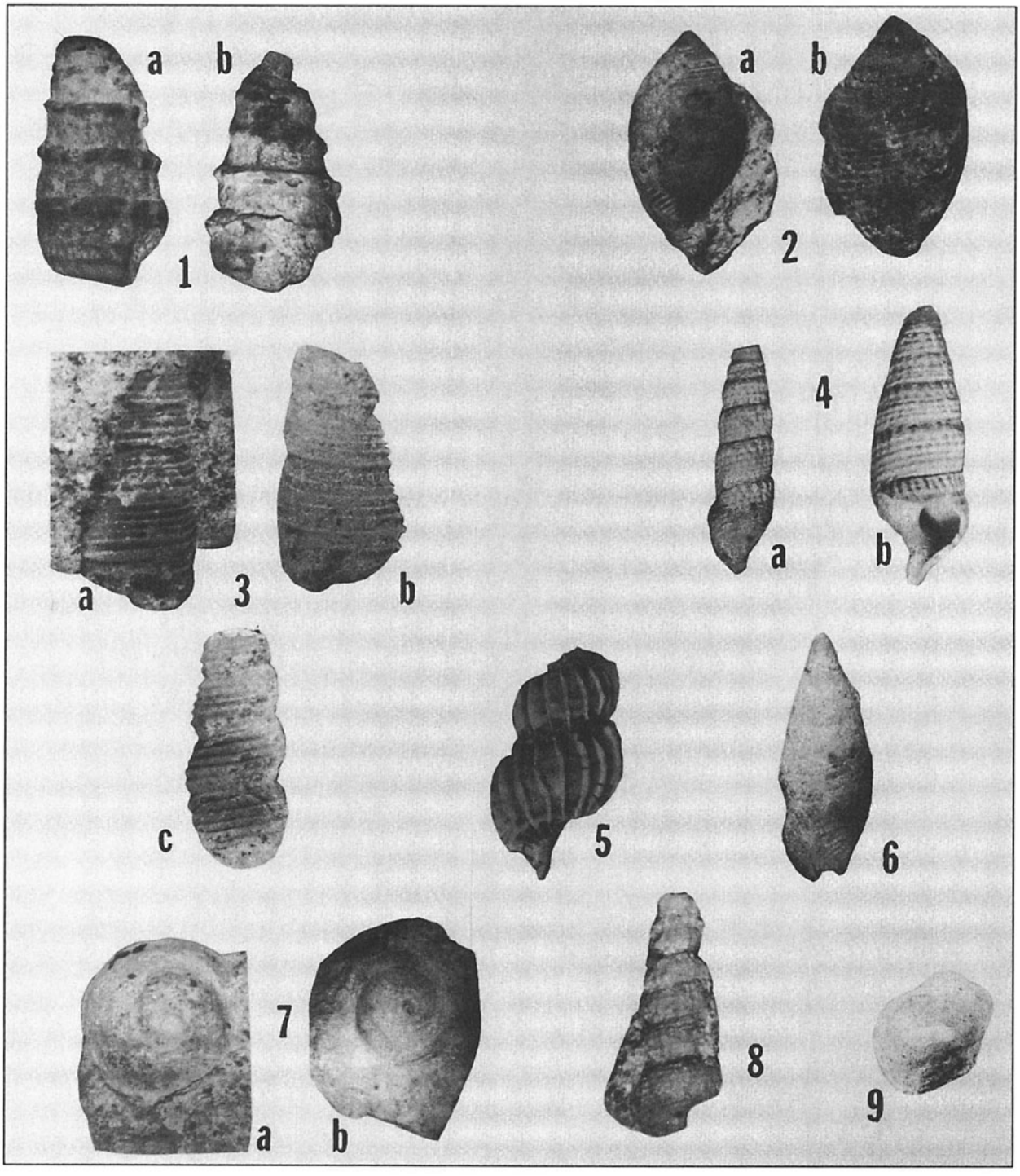


Figure 3. Illustrations of marine gastropods from the "SB Hill" and "Slump Hill" sections. (1a–1b) *Tympanotonos* aff. *papalis* (x1.35); (2a–2b) *Acteon moodyi* (x3.6); (3a–3c) *Turritella uvasana infera* (x2.7); (4a–4b) Cerithidae? (a. x1.8, b. x3.6); (5) *Cirsotrema* sp. (x3.6); (6) *Ectinochilus* cf. (*Macilentos*) *macilentus* (x1.1); (7a–7b) *Calyptraea* cf. *diegoana* (x0.9); (8) *Cerithium excelsum* (x7.2); (9) *Polinices* (*Euspira*) *nuciformis* (x0.45).

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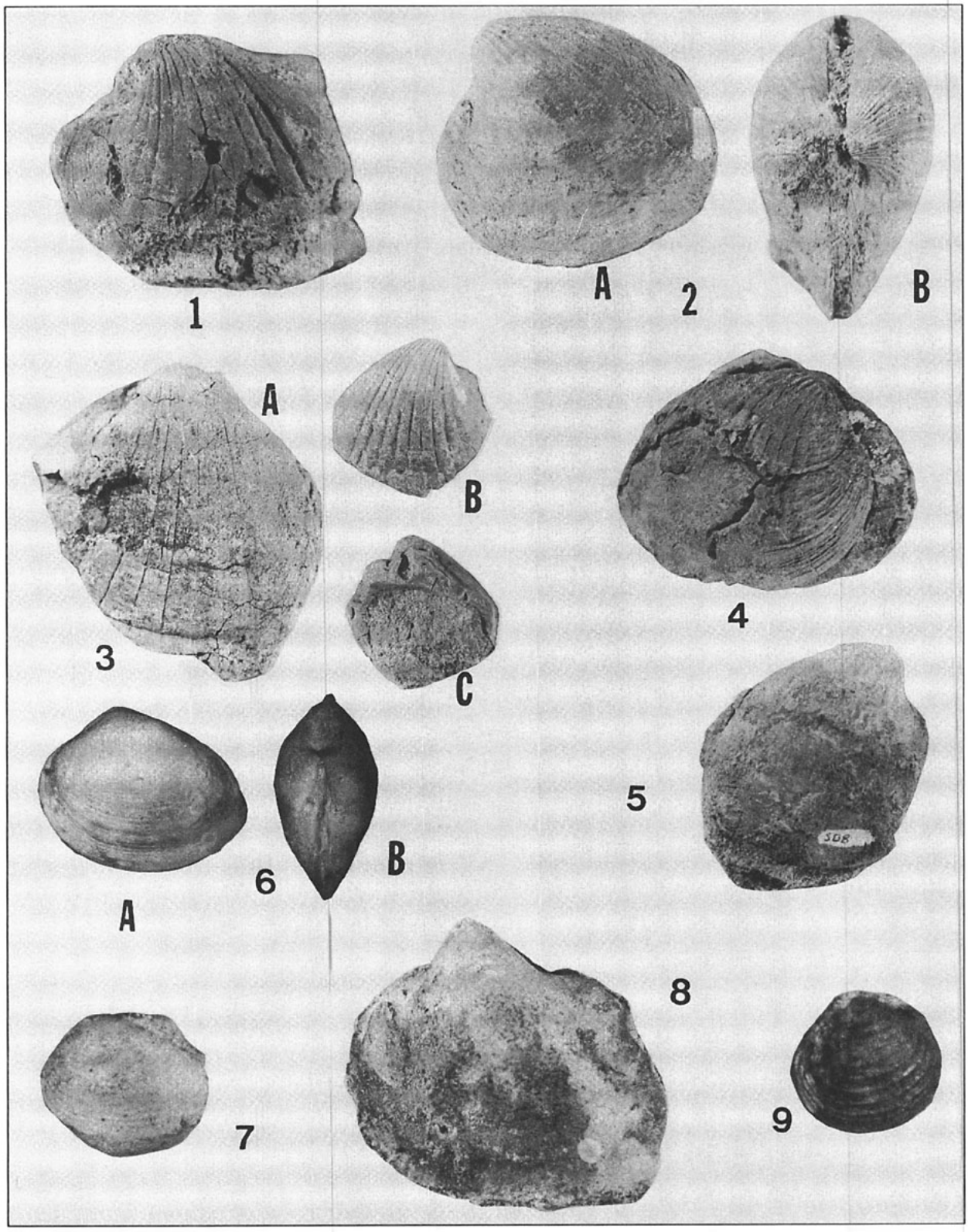


Figure 4. Illustrations of marine pelecypods from the "SB Hill" and "Slump Hill" sections. (1) *Venericardia* (*Pacifor*) cf. *aragonia* (right valve, x0.9); (2) *V.* (*P.*) *diabloensis* (a. right valve; b. hinge view; x0.6); (3) *V.* (*P.*) *aragonia joaquinensis* (a. right valve, x0.9; b–c. left valve, x0.55); (4) *V.* (*P.*) *durhami* (left valve, x0.45); (5) *V.* (*P.*) *aragonia lutmani* (left valve, x0.45); (6) *Macrocallista* sp. nov.? (a. right valve; b. hingeview; x.2.25); (7) *Macrocallista* (*Costacallista*) *domengenica capayana* (right valve; x1.8); (8) *Crassatella* sp. (steinkern, x0.55); (9) *Crassatella* cf. *meganosensis* (x3.6).

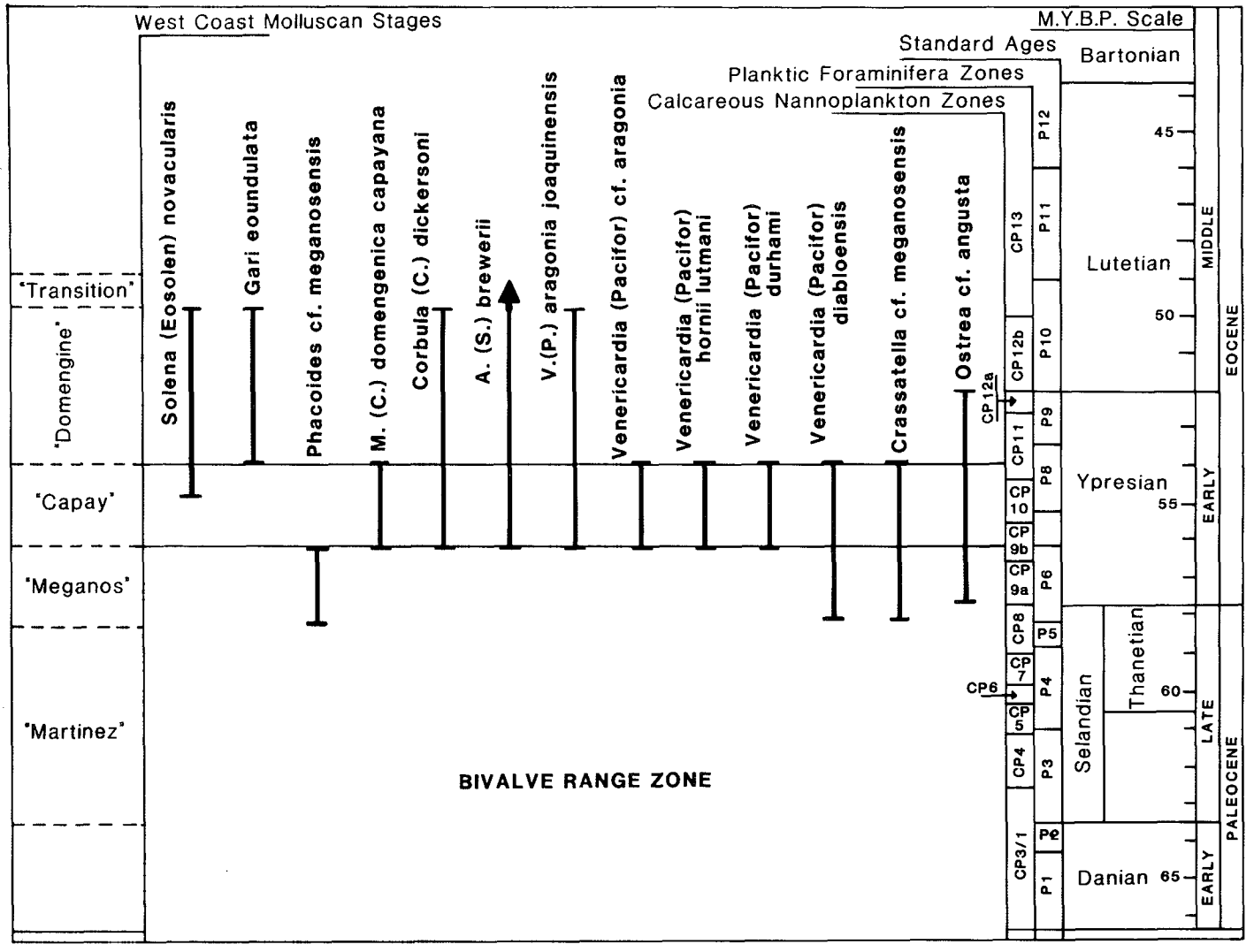


Figure 5. Standard biostratigraphic range zones of bivalve taxa present in the "SB Hill" and "Slump Hill" sections. Zonation and ranges after Saul (1983a and 1983b), Squires (1984), Keen and Bentson (1944), and Givens (1974).

zian Stage) first appearance, and four of the remaining five taxa first appear in Bulitian (early Eocene) or younger stages. The "Slump Hill" section contained two benthic foraminifera that were not useful for temporal correlation: poorly preserved *Gavelinella* sp. and the long-lived species *Globulina gibba*. Although the assemblages used to define benthic foraminiferal stages have been shown to be time transgressive in some cases (for example, McDougall, 1980; see related suggestions in Schenck and Kleinpell, 1936; Mallory, 1959), detailed studies of sequences containing benthic foraminiferal assemblages and independent temporal indicators (calcareous nannoplankton and planktonic foraminiferal assemblages) permit relatively precise temporal correlations of the benthic foraminiferal stages (McDougall, 1980; Poore, 1980; Warren, 1983). The Ynezian to Narizian stages range from the latest Paleocene to middle Eocene; the ranges of all six taxa best support a Penutian or Ulatisian Stage assignment, and an early Eocene (Ypresian) age (Warren, 1983) for the assemblage and the associated strata.

Ostracodes

The marine strata of this study have produced the ostracode *Buntonia* (F. Swain, 1986, personal commun.). *Buntonia* has an Eocene first appearance in other areas.

Geochronology

The most reliable marine biochronologies for global temporal correlation are those based on planktonic foraminifera or calcareous nannoplankton. The Baja California sequences do not contain either planktonic foraminifera or calcareous nannoplankton. Therefore, we use a correlation network in which (1) the Baja California molluscan and benthic foraminiferal assemblages are correlated to North American West Coast Stages; (2) the North American West Coast molluscan and benthic foraminiferal stages have been directly correlated to standard, global planktonic foraminiferal and calcareous nannoplankton biochronologies; and (3) the planktonic foraminiferal and calcareous nannoplankton biochronologies have been correlated with the standard magnetic polarity chronology. This correlation network ultimately permits correlation of the Baja California sequence to standard global geochronologies (for example,

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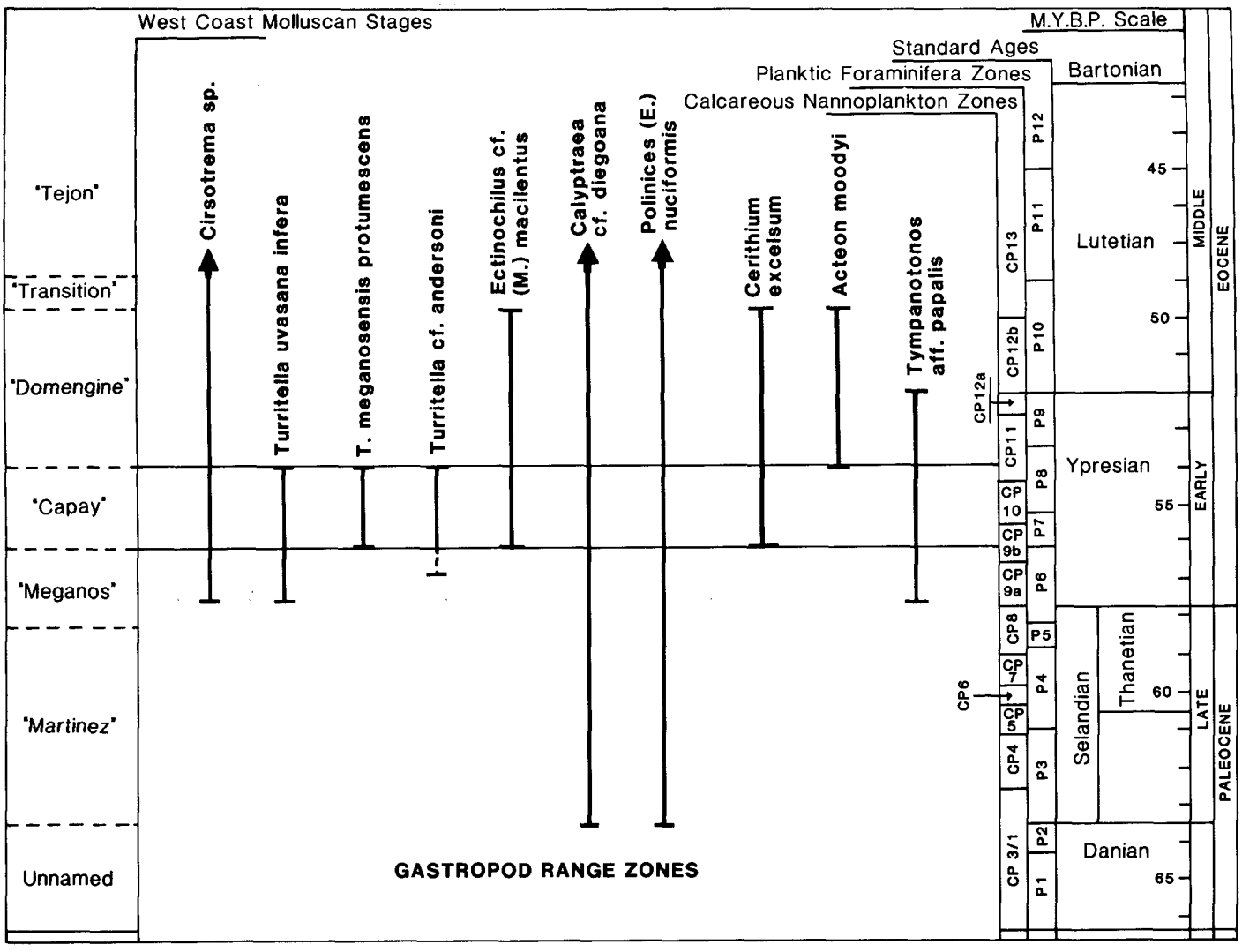


Figure 6. Standard biostratigraphic range zones of gastropod taxa present in the "SB Hill" and "Slump Hill" sections. Zonation and ranges as in Figure 3.

Berggren and others, 1985). Using the geochronology of Berggren and others (1985) and the correlations of Saul (1983a, 1983b), Squires (1984), Poore (1980), and Warren (1983), the marine biostratigraphies indicate a middle Ypresian (early Eocene) age for the strata of this study. A more precise age estimate would indicate approximate correlation with early P7 to late P8 (planktonic foraminiferal zones; Berggren, 1969) and mid-CP9b to early CP11 (calcareous nannoplankton zones; Bukry, 1973, 1975; Okada and Bukry, 1980).

Biogeography

Warm-water, early Cenozoic faunas had a greater geographic range than modern faunas because of warmer climatic conditions and a reduced global climatic gradient during the early Cenozoic (Addicott, 1970). This is emphasized by the ability to recognize the early Cenozoic faunal zones of Clark and Vokes (1936) from the Gulf of Alaska to Baja California, Mexico. Our molluscan assemblage clearly represents a tropical to subtropical assemblage (Table 4). The occurrence of taxa in this assemblage that have an affinity with Paris Basin taxa is not particularly surprising. Vokes (1939), Givens and Kennedy (1976), and Squires (1984) reported that a small percentage of taxa in Californian early to middle Eocene assemblages bore a strong resemblance to Paris Basin forms. This clearly is applicable to faunas from Baja as well.

Palmer (1967) noted that many areas of North America have Paleogene molluscan faunas of Tethyan affinities with taxa comparable to Paris Basin taxa. It has been postulated that seaways existed across what is now Central America from Jurassic through Eocene, even though the Tethyan seaway began closing during the Paleogene (see Berggren and Hollister,

TABLE 3. BENTHIC FORAMINIFERA IN "SB HILL" LOCALITY

Taxa	Benthic Foram Stage
Elphidium californicum	Bulitian through Ulatisian
Quinqueloculina cf. *josephina*	Ulatisian
Quinqueloculina triangularis	Ynezian through Penutian
Cibicides cf. *cushmani*	Bulitian through Narizian
Discorbis cf. *coalingensis*	Ulatisian
Amphistegina sp.	Upper Ynezian through Narizian

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TABLE 4. CLIMATIC/FAUNAL REGION DISTRIBUTION OF EXTANT REPRESENTATIVES OF MARINE GENERA OCCURRING IN THE STUDY AREA

Genus	Boreal	Warm temp.	Tropical
Bivalvia			
Acanthocardia	x	M	
Corbula (Carycorbula)	x	x	M
*Crassatella**		x	M
Gari	x	x	M
Macrocallista		x	M
Ostrea	x	x	M
Phacoides	?	x	M
Pholodomya		x	M
Solena			M
Teredo†	x	M	x
Venericardia (Pacifor)		x	M
Gastropoda			
Acteon		x	M
Calyptraea	x	x	M
Cerithidea		x	M
Cerithium		x	M
Cirsotrema		x	M
*Ectinochilus**	?	x	M
Polinices (Euspira)	x	x	M
Turritella		x	M
Crustacea§			
Callianassa	?	x	x
Cnidaria**			
Goniopora			M

*Extinct genera, ranges inferred from paleobiogeographic distribution.
x, present; M, most frequently reported occurrence.
†Trace fossils, possibly made by the genus *Teredo*; SB Hill Loc.
§Claws probably assignable to *Callianassa*; in both the SB and SH locs.
**Coral fragments, probably assignable to *Goniopora*; SB loc.
Note: based on information from Givens (1974), Abbott (1974), Keen and Coan (1974), Saul (1983), and Squires (1984). Table after Squires (1984).

1974; Kennett, 1982; Squires, 1984). This study, and the work of others, indicates that the Tethyan seaway remained open until at least the middle to late Eocene. Although migration of some taxa appears to have been from east to west during the Paleogene, there is evidence that migration during the Jurassic was from west to east (Bayer and McGhee, 1985); these differences probably reflect changes in oceanic circulation during the fragmentation of Pangea.

Our faunal collection of Baja California early Eocene molluscans shows no obvious similarity to South American or Gulf Coast faunas. Clark and Vokes (1936) and R. Squires (1985, personal commun.) noted that other Paleogene generic assemblages of the west coast of North America were quite different from those of the Gulf Coast. The reasons for these differences are unclear.

STRATIGRAPHIC RELATIONS OF THE MARINE AND TERRESTRIAL SECTIONS

The two marine sections are areally separated by ~3 km, and the marine ("SB Hill") and terrestrial ("Lomas las Tetas de Cabra/GTXX" and "Younger World") sections are areally separated by <15 km (Fig. 2). Discontinuous exposures between the marine and terrestrial sections (erosion and original discontinuous deposition controlled by pre-existing topographic highs, such as Mesozoic igneous and metamorphic hills) make it difficult to directly trace stratigraphic continuity between the sections. The almost horizontal nature of the strata in both areas, however, and no evidence of post-depositional faulting between the two sections permit approximate stratigraphic correlations. These geometric relationships indicate approximate stratigraphic equivalence between the "SB Hill" and "Lomas las Tetas de Cabra"/"Younger World" sections; the "Slump Hill" section is slightly older than the "SB Hill" section, although the two sections significantly overlap stratigraphically. Further, lagoonal to nonmarine strata directly intertongue with "Capay" Stage marine strata in the area between the "SB Hill" and "Lomas las Tetas de Cabra"/"Younger World" sections. These lagoonal to nonmarine tongues contain numerous *Celtis* seeds, edentulous mammal jaws, and a single specimen of *Hyracotherium* (reported by Morris, 1966). Abundant *Celtis* seeds and the diagnostic Wasatchian taxon *Hyracotherium* are found in the "Lomas las Tetas de Cabra" section. Abundant occurrences of *Celtis* are characteristic of Wasatchian strata elsewhere in North America (B. Tiffney, 1985, personal commun.).

PALEOMAGNETISM

Field and Laboratory Studies

(A) Field Sampling Methods. At least three individually oriented sample blocks were collected at each sampling site for paleomagnetic analyses. All paleomagnetic sampling sites were located directly within measured stratigraphic sections. Sampling in the paleomagnetic sections included the (1) "Las Tetas de Cabra" section (LTC/GTXX), 23 sites, 58.9 m; (2) "Younger World" (YW) section, 18 sites, 51 m; and (3) "Slump Hill" section (SH), 13 sites, 25 m. Correction for bedding tilt involved rotation of the bedding to horizontal about the strike; we think that this simple tectonic correction is reasonable, given the simple structure of Cenozoic rocks in this region.

Laboratory preparation entailed cutting or grinding blocks into ~2.5-cm cubes; all samples were hand sanded before measurement to remove surface impurities. Most blocks produced several subsamples, which permitted complementary demagnetization studies (AF and thermal) for individual samples.

(B) Pilot Demagnetization Studies. All paleomagnetic analyses were performed in the Rutgers University or Field Museum paleomagnetics laboratories, using the Molspin Minispin spinner magnetometer, Molspin Minisep magnetic susceptibility system, and Schonstedt TSD-1 thermal demagnetizer and GSD-1 A.C. demagnetizer.

Detailed, step-wise alternating field (to 90 mT) and thermal demagnetization (675 °C) studies were performed on one sample from each sampling site. Representative vector-demagnetization plots for both terrestrial (LTC 4B1 and 6B1) and marine (SH 4B and 13A1) sediments are shown in Figure 7. These demagnetization studies show that characteristic remanences are isolated at very low demagnetization fields and temperatures, with many sample NRM's preserving the characteristic remanence directions. Both alternating field and thermal demagnetization procedures effectively isolate the characteristic remanences. After removal of small, or negligible, unstable viscous magnetizations, most sample magnetizations exhibit univectorial decay toward the origin. Some sample demagnetization curves, however, show removal of small magnetization components that are approximately antiparallel to the characteristic magnetizations (for example, SH 4B). These components probably represent present-field overprints, as they are oriented to the north and down. Most sample magnetizations are of very low intensity following demagnetization above 600 °C, although many samples in the terrestrial "Lomas las Tetas de Cabra" and "Younger World" sections do retain a stable, low-intensity magnetization component above 600 °C. Directions of the high-temperature magnetization components are the same as the low-temperature magnetization component.

(C) Demagnetization and Statistical Analyses. On the basis of magnetization vector behavior during pilot-demagnetization studies, remaining samples from each site were demagnetized in several thermal demagnetization steps. Characteristic remanence directions were determined for each sample (stable end point or principal component analysis). Site-mean directions, and site VGP's, were calculated from the sample directions (Fisher, 1953).

Site-mean directions for both sections are given in Table 5. It is obvious that individual site means within each section are tightly clustered and belong to a single population, as reflected in the high precision parameters (k). Inclinations for the section means are very close to that expected for the present site latitude (±47.8°). Section-mean declinations, however, deviate significantly from an expected value of 180° (for reversed-polarity sites). These deviations indicate ~15°–20° of counterclockwise rotation of the sampling area since the early Eocene. Stereonet plots (Fig. 8) of the section-mean directions show overlap of the circles of confidence, indicating that the section means are not statistically significantly different. These plots graphically illustrate the 15°–20° of counterclockwise rotation of all of the sections. Paleolatitudes calculated from the section-mean directions (Table 5, using the corrected formula presented in

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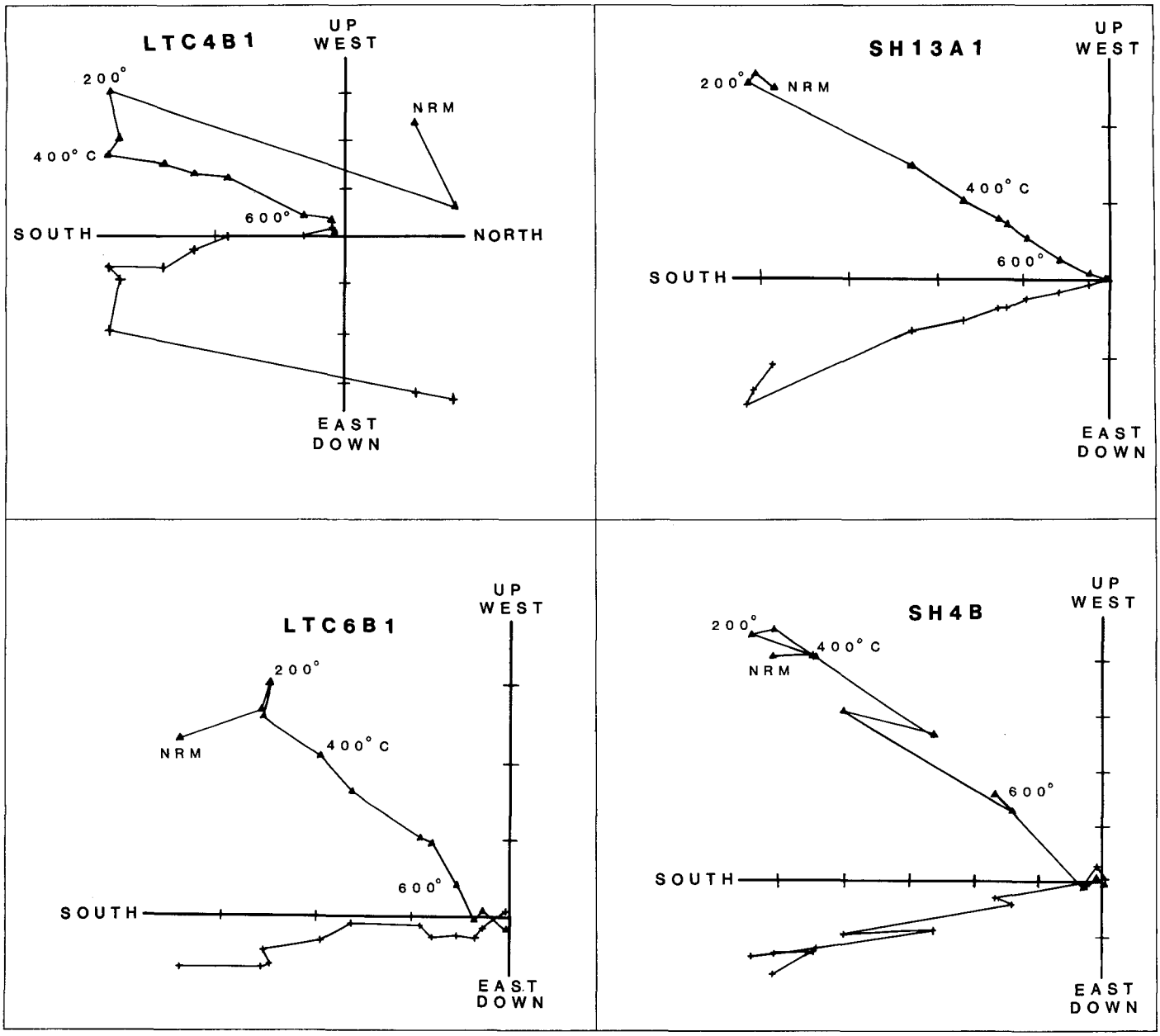


Figure 7. Vector demagnetization diagrams for representative pilot samples from the "Lomas las Tetas de Cabra" (LTC) and "Slump Hill" (SH) paleomagnetic sections. Progressive thermal demagnetization from NRM to 650 °C. Crosses, horizontal component of magnetization vector; triangles, vertical component of vector. Scale divisions are in $10^{-3}/4\pi$. Amps/meter.

Haile, 1975) are almost precisely the same as the present site latitude.

The reliability classification of Opdyke and others (1977) was used to evaluate the reliability of site magnetization directions for polarity determinations. Class I reliability sites are those sites that have statistically significant magnetization directions, *after* thermal demagnetization treatment (modified from Opdyke and others, 1977; using Watson's 1956a and 1956b statistical tests).

(D) Determination of Remanence Carriers. The strata in this study are almost flat lying; lack of tilting or deformation precludes application of a fold-test for magnetic stability. All characteristic magnetizations are of reversed polarity (see below), which also precludes a reversals test. Therefore, properties of the magnetization carriers provide the best support for concluding that the characteristic magnetizations are primary remanences. First, the reversed polarity of the characteristic magnetization indicates that it is not a secondary magnetization acquired in the present-day field during recent uplift, weathering, or ground-water alteration. Second, a single magnetization carried by magnetite is present at most sites in the marine sections; this magnetite remanence is almost certainly primary. Third, the concordance of magnetization directions between the remanences in the marine (magnetite-remanence carrier) and the terrestrial (both magnetite- and hematite-remanence carriers) sections indicates that the characteristic remanence in the terrestrial section is also primary. Further, directional concordance between the moderate and high blocking temperature magnetization components in the terrestrial section indicates that the hematitic remanence probably was acquired syndepositionally, or shortly after

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TABLE 5. PALEOMAGNETIC SITE-MEAN DIRECTIONS

Site #	DEC	INC	N	K	CofC	R	PLNG	PLAT
Lomas las Tetas de Cabra Section (LTC)/GTXX Section (GTXX)								
GTXX 1	188.0	-49.1	3	20.33	28.1	2.902	142.8	-82.9
GTXX 2*	183.0	-39.3	3	3.27	83.7	2.388	222.2	-83.1
GTXX 3	206.9	-63.4	4	48.35	13.4	3.938	111.2	-63.2
GTXX 4*	25.5	-64.3	3	1.08	>90	1.151	48.2	11.8
GTXX 5	158.3	-49.1	3	113.91	11.6	2.982	345.5	-71.1
GTXX 6	187.2	-54.4	3	30.49	22.7	2.934	108.2	-81.2
LTC 1	175.2	-42.8	4	60.05	12.0	3.950	295.9	-84.3
LTC 2	172.1	-51.9	3	90.61	13.0	2.980	7.9	-82.2
LTC 3	159.9	-46.8	4	40.69	14.6	3.926	339.0	-72.3
LTC 4	164.7	-47.8	3	38.48	20.2	2.948	340.9	-76.6
LTC 5	169.6	-36.3	3	17.25	30.6	2.884	296.6	-77.3
LTC 9*	178.2	-33.8	2	12.06	79.7	1.917	255.7	-79.8
LTC 6	179.2	-43.8	3	87.75	13.2	2.977	259.6	86.9
LTC 7	173.7	-52.2	3	304.21	7.1	2.993	15.4	-83.1
LTC 8	176.0	-70.8	4	5.21	44.6	3.424	60.9	-63.3
LTC 10	145.4	-36.8	3	156.15	9.9	2.987	329.4	-57.6
LTC 11	144.4	-40.0	3	14.99	33.0	2.867	334.0	-57.5
LTC 12	135.8	-28.7	3	342.97	6.7	2.994	327.6	-47.2
LTC 13	181.0	-52.4	3	136.21	10.6	2.985	76.8	-85.5
LTC 14	178.7	-70.0	3	26.11	24.6	2.923	64.2	-64.6
LTC 15	154.1	-47.1	3	39.37	19.9	2.949	341.5	-67.3
LTC 16	165.5	-51.4	3	42.45	19.2	2.953	355.1	-77.0
LTC 17	176.7	-41.7	3	12.43	36.6	2.839	279.6	-84.5
Mean	167.5	-50.0	16	30.25	6.8	15.504	343.1	-76.2
"Slump Hill" Section (SH)								
SH 14	131.4	-29.5	3	5.72	57.2	2.650	330.9	-43.6
SH 15*	124.2	-7.3	3	4.33	68.4	2.538	321.6	-31.6
SH 1	174.0	4.5	3	5.67	57.5	2.647	257.7	-58.7
SH 2	143.3	-33.4	3	136.13	10.6	2.985	327.2	-55.0
SH 3	155.8	-48.7	3	51.42	17.4	2.961	344.9	-68.9
SH 4	156.1	-41.7	4	71.62	10.9	3.958	329.7	-68.2
SH 5	143.4	-48.5	4	6.04	40.8	3.503	346.9	-58.1
SH 6	153.3	-49.4	4	24.81	18.8	3.879	346.9	-66.7
SH 7	155.3	-41.3	4	208.96	6.4	3.986	346.7	-68.5
SH 8	155.0	-47.7	5	451.63	3.6	4.991	342.8	-68.1
SH 9*	180.9	-48.6	3	4.62	65.5	2.567	102.4	-88.7
SH 16	160.0	-14.4	3	152.00	10.0	2.987	291.6	-61.6
SH 12	155.8	-38.8	4	16.49	23.3	3.818	324.7	-67.2
Mean 1	153.5	-36.2	11	18.09	11.0	10.447	322.7	-64.5
Mean 2	153.3	-40.7	9	45.79	7.7	8.825	329.8	-65.5
Younger World Section (YW)								
YW 1	165.2	-41.2	4	45.94	13.7	3.935	318.9	-75.8
YW 2	153.3	-32.6	4	26.30	18.3	3.886	317.9	-63.2
YW A3	168.0	-41.6	3	65.53	15.4	2.969	315.3	-78.3
YW B3	152.1	-50.0	3	48.77	17.9	2.959	348.1	-65.7
YW C3	162.5	-56.0	3	42.47	19.2	2.953	8.9	-73.3
YW 4	170.0	-53.8	3	27.55	24.0	2.927	12.5	-79.7
YW 5	195.1	-57.4	3	80.78	13.8	2.975	115.3	-74.3
YW 6	149.1	-45.9	3	5.50	58.6	2.637	340.7	-62.8
YW 7	23.9	-70.2	5	2.69	58.1	3.512	52.3	4.6
YW 8	151.5	-51.7	3	233.78	8.1	2.991	351.8	-65.2
YW 9	161.6	-36.9	4	103.49	9.1	3.971	314.5	-71.5
YW 10*	160.2	-64.1	3	4.58	65.9	2.564	29.3	-66.8
YW 11	166.7	-51.4	3	12.23	36.9	2.837	356.1	-78.0
YW 12	171.4	-60.7	4	18.10	22.2	3.834	40.2	-75.2
YW 13	182.9	-46.7	3	23.85	25.8	2.916	169.6	-87.4
YW 14	205.0	-62.4	4	23.81	19.2	3.874	112.2	-65.0
YW 15	177.6	-45.9	4	28.18	17.6	3.894	304.9	-87.5
YW 16	141.1	-56.4	4	4.86	46.6	3.383	0.7	-56.5
YW 17	170.7	-54.4	4	23.50	19.4	3.872	17.1	-79.9
YW 18*	218.5	-61.9	3	1.61	>90	1.756	120.3	-56.0
Mean	165.0	-53.4	18	20.65	7.8	17.177	2.3	-76.1
Composite section mean								
	161.4	-48.2	3	93.39	12.8	2.979	342.7	-73.7
Composite individual site mean								
	162.5	-49.3	48	23.04	4.4	45.960	346.1	-74.7

Mean, section mean; Mean 1 includes SH1 and SH14, Mean 2 excludes SH1 and SH14 (marginally significant); Composite section mean, mean of LTC/GTXX, YW, and SH (Mean 2) section means; Composite individual site mean, mean of all statistically significant sites; DEC, site mean declination; INC, site mean inclination; N, number of samples; K, kappa, magnetic precision parameter; CofC, angular radius of the 95% cone of confidence about the observed mean; R, resultant vector (giving unit values to each N vector); PLNG, PLAT, paleopole longitude and latitude calculated from the mean direction; *, mean direction not statistically significant (Watson, 1956b).

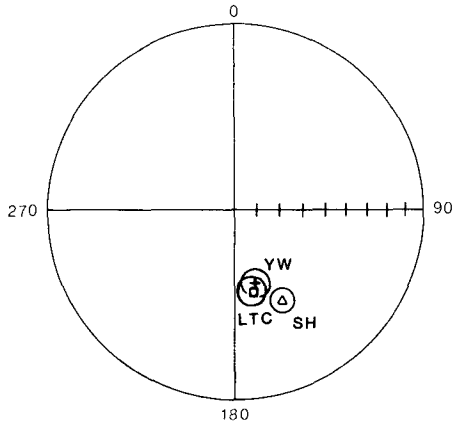


Figure 8. Stereonet plots of mean directions of magnetization for the "Lomas las Tetas de Cabra/GTXX" (LTC), "Younger World" (YW), and "Slump Hill" (SH) paleomagnetic sections. Section means are of reversed polarity (negative inclinations, plot on upper hemisphere).

deposition. Fourth, concordant remanence directions between the radically different lithologies of the marine and terrestrial strata would be difficult to explain through simple acquisition of a secondary remanence. Fifth, some tectonic rotation, indicated by deviation of the declinations from expected Cenozoic values, with no indication of recent local tectonic rotation, argues against a recent secondary origin for the characteristic remanences.

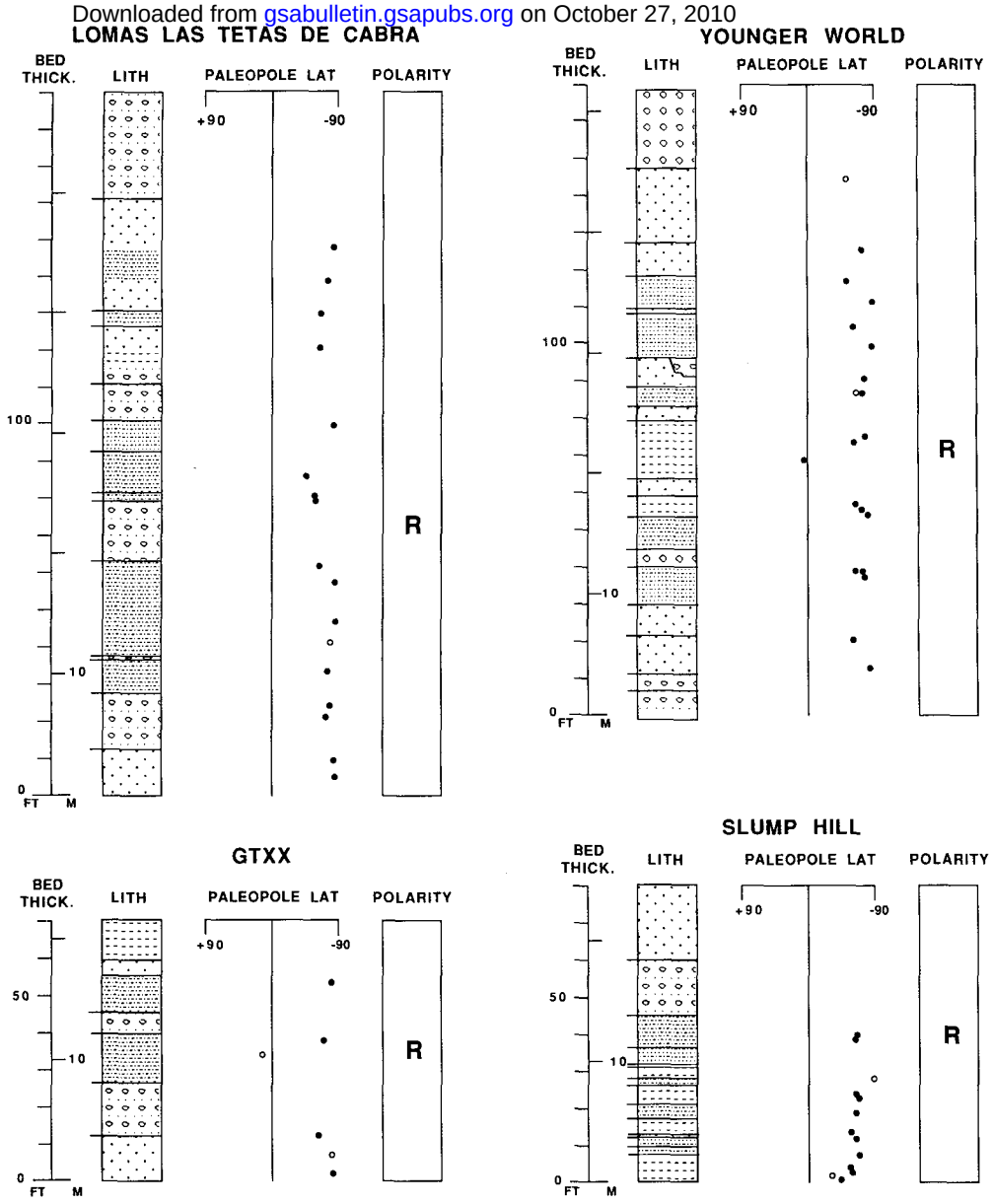
Champion and others (1986) suggested that some of their middle Eocene samples from San Miguel Island, California, acquired a secondary thermal magnetization from Miocene volcanic activity. This thermal overprint affected 55 of the total 117 middle Eocene cores and could not be completely removed by either A.F. or thermal demagnetization. A similar Miocene thermal overprinting is unlikely in our Eocene samples because of the remanence characteristics of these samples and the lack of extensive Miocene igneous rocks in the study area.

Magnetostratigraphy

The magnetic polarity stratigraphies, constructed from the site-mean magnetization information, for the terrestrial "Lomas las Tetas de Cabra" and "Younger World," and for the marine "Slump Hill" sections are shown in Figure 9. In all three sequences, all sites are of reversed magnetic polarity. On the basis of the intertonguing of the marine and terrestrial strata in the "SB Hill" section area, we tentatively conclude that these two reversed polarity intervals are contemporaneous. Correlation of the magnetostratigraphies to a standard geomagnetic polarity time scale (GMPTS) is impossible on the basis of the pattern of magnetic polarity alone, and there are no radioisotopic dates from these sections. As described above, however, establishment of a correlation network permits correlation of the magnetostratigraphies to the GMPTS. The validity of these temporal correlations depends on the reliability of each component of the correlation network.

The presence of restricted "Capay" Stage molluscan assemblages and Ynezian to Narizian benthic foraminifera requires temporal correla-

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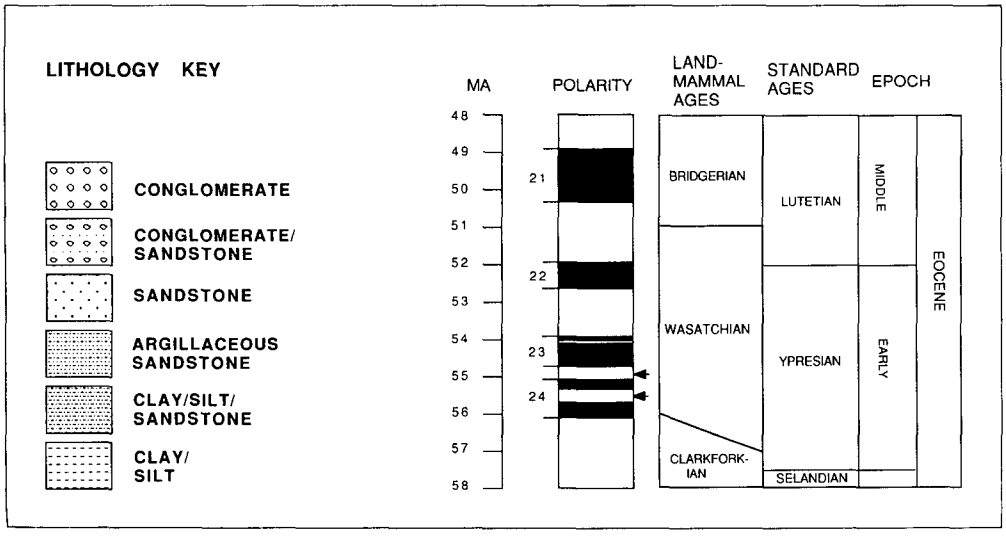


Figure 9. Magnetic polarity stratigraphy. "Lomas las Tetas de Cabra" (LTC), "GTXX" (GTXX), "Younger World" (YW), and "Slump Hill" (SH) sections; plotting bedding thickness (in feet and meters), lithology, paleopole latitudes calculated from site mean magnetizations, and inferred magnetic polarity zonation; magnetic polarity time scale from Berggren and others (1985); large arrows indicate two possible correlations (determined from biochronology) for the LTC/GTXX/YW/SH reversed magnetozone.

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tion of these strata to the Ypresian. The molluscan and benthic foraminiferal assemblages are best correlated to the middle Ypresian planktonic foraminiferal biozones P7 to late P8 and calcareous nannoplankton biozones middle CP9b to early CP11 (Berggren and others, 1985). In the geochronology of Berggren and others (1985), these biozones correlate with magnetic polarity Chrons C23N to C24N. Therefore, the reversed polarity interval in both the terrestrial and marine sections most likely correlates in time with Chron C23R or the reversed polarity interval within Chron C24N. Correlation to either the younger Chron C22R or the older Chron C24R would require significant asynchrony of the molluscan and foraminiferal assemblages between Baja California and other parts of the North American West Coast. There is no evidence to support this unlikely possibility of faunal asynchrony.

Paleolatitudes

A series of recent studies use paleomagnetism to determine the plate-tectonic motion of the Baja California peninsula, relative to the North American craton, during the Mesozoic and Cenozoic.

Marshall and others (1979) suggested that Baja was 10° farther south in the Miocene, relative to its present position.

TABLE 6. CALCULATED PALEOPOLES/PALEOLATITUDES FOR THE BAJA SECTIONS

Section	Paleopole longitude	Paleopole latitude	Site paleolatitude
LTC/GTXX	170.4	79.0	30.8°N
YW	182.3	76.1	33.9°N
SH	149.8	65.5	23.4°N
Sections combined	162.7	73.7	29.2°N
Sites combined	166.1	74.7	30.2°N

Hagstrum and others (1985) invoked northward translation of 11° and clockwise rotation of Baja since the Cretaceous. They believed the northward translation was completed by the late Miocene (see Fig. 10).

Morris and others (1986) considered the Baja peninsula as part of the Peninsular Ranges Terrane (PRT). The PRT had a Late Cretaceous paleolatitude 6° farther south than at present, whereas the North American craton was ~9° farther north, yielding a latitudinal offset of 15°. Morris and others (1986) considered the PRT as part of the stable North American block from 160–15 Ma, although Baja and the PRT was located 15° farther south along the margin of Central America. Therefore, the entire 15° of northward translation of Baja occurred from 15 Ma to the present (see Fig. 10).

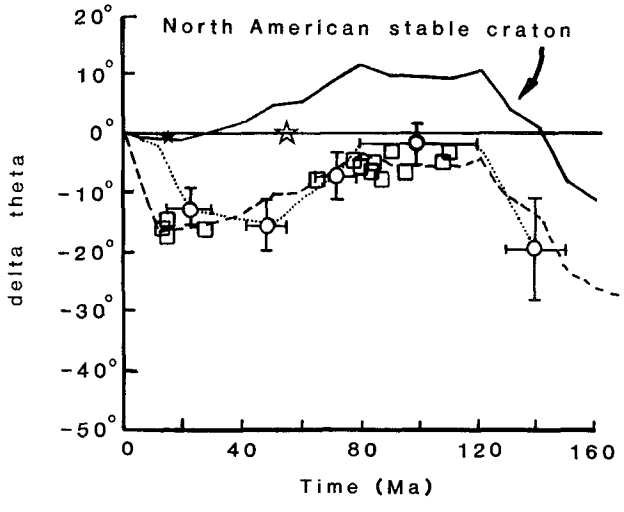


Figure 10. Normalized paleolatitudes (delta theta; paleolatitude minus present latitude) for Baja California, plotted versus time (after Morris and others, 1986). The North American stable craton reference curve derived from Irving and Irving (1982) by Morris and others (1986); squares, Peninsular Ranges Terrane (PRT) data presented by Morris and others (1986); dashed-line curve was originally drawn by Morris and others (1986) to generally fit the PRT data and parallel the North American curve (except in the Miocene to Recent); circles, Baja-Borderland Allochthon (BBA) data of Champion and others (1986), associated error bars show temporal uncertainty and paleomagnetic error estimates, (note: the 0–80 Ma data were not derived from the Santa Ana–Cortez terrane blocks of the Baja California Peninsula mainland); dotted line, BBA paleolatitude path proposed by Champion and others (1986); small solid star, Miocene (10–20 Ma) data of Hagstrum and others (1987) from mainland Baja California; large open star, early Eocene data of this study from mainland Baja California.

Champion and others (1986) investigated the paleomagnetism of Cretaceous and middle Eocene rocks of San Miguel Island, California, which they considered western lithofacies of rocks cropping out on the Baja California peninsula and part of the "Baja-Borderland Allochthon." With respect to the expected directions for the present site positions, Upper Cretaceous rocks were 19° shallower and rotated 50° clockwise (paleolatitude of 26°), and middle Eocene rocks were 24° shallower and rotated 79° clockwise (paleolatitude of 17°). When that was combined with other available paleomagnetic information, Champion and others (1986) concluded that the motion of the Baja Peninsula and Peninsular Ranges of southern California mimicked that of the North American craton from the Late Cretaceous to the late Eocene; Baja arrived at its pre–Gulf of California position sometime between the Oligocene and earliest Miocene.

In a study of Miocene volcanics, Hagstrum and others (1987) concluded that Baja was in its pre–Gulf of California position from at least the early Miocene (see Fig. 10), contrary to the results of Marshall and others (1979). Hagstrum and others (1987) suggested 9° of northward translation and 24° of clockwise rotation of Baja during the early Cenozoic, possibly during the major plate reorganization between 45 and 34 Ma.

It is clear from this brief summary of previous studies that (1) the Baja peninsula has been translated to the north ~10°–15°, relative to the North American craton, since the Late Cretaceous; (2) only one previous study (Champion and others, 1986) constrained the motion of Baja during the early Cenozoic (middle Eocene), and their results must be analyzed cautiously because of weak, relatively unstable magnetizations, some of which have been thermally overprinted; (3) previous studies are mutually contradictory regarding the timing of northward translation; several studies (Hagstrum and others, 1985, 1987; Champion and others, 1986) stated that translation was completed prior to the Miocene (probably during the early Cenozoic), whereas Morris and others (1986) concluded that translation was entirely post-Miocene.

Comparison of our paleomagnetic results to those of previous studies reveals several important conclusions. First, the age of the strata analyzed in our study is very well determined as middle early Eocene. Second, remanence characteristics are very stable; it is easy to isolate a univectorial characteristic remanence that almost certainly is primary. Third, section-mean directions indicate an early Eocene paleolatitude for the Baja California peninsula that is very close to its present position. This early Eocene

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paleolatitude is only a few degrees from the position predicted for Baja by the pole path for cratonic North America (Fig. 10).

The results of our study indicate that there has been no significant post–early Eocene northward translation of Baja California. If Baja was 10°–15° farther south during the Late Cretaceous, relative to the North American craton, then northward translation of the Baja peninsula must have been completed before the middle early Eocene.

DISCUSSION

Wasatchian faunas have been used extensively for temporal correlation between North America and Europe. Such precise correlations between widely geographically separated areas are possible because of a continuous terrestrial connection between North America and Europe across the North Atlantic during the early Eocene (McKenna, 1983, 1984). These correlations have led to the conclusion that the Paleocene/Eocene boundary lies within the Clarkforkian (Rose, 1980, 1981; Gingerich and others, 1980), although it may lie nearer to the beginning of the Wasatchian (Wing, 1984; Berggren and others, 1985). Butler and others (1981) and Rapp and others (1983) correlated terrestrial magnetostratigraphies (of sections containing Clarkforkian and Wasatchian mammal faunas) to standard geochronologies, assuming that the Paleocene/Eocene boundary lies within the Clarkforkian. These correlations indicated that the Clarkforkian extended from the youngest part of Chron C25N to medial or late Chron C24R, and early Wasatchian faunas (sections containing later Wasatchian faunas have not yet been sampled for paleomagnetic stratigraphy) spanned medial or late Chron C24R through C24N. Tauxe and Clark (1987) indicated that Ellesmere Island Wasatchian faunas occurred within a reversed polarity interval that probably correlated with Chron C24R, consistent with the results of Butler and others (1981) and Rapp and others (1983). No previous study, however, has directly correlated Clarkforkian or Wasatchian faunas to standard, global marine biochronologies.

The Baja "Lomas las Tetas de Cabra" sequence provides the first direct correlation between earliest Cenozoic marine and terrestrial biochronologies in North America. The only other early Paleogene sequence for which such a direct correlation exists is the middle Eocene of the San Diego area (see Flynn, 1986; Golz and Lillegraven, 1977). The marine biochronologies (and the dependent magnetostratigraphic correlation) indicate a middle Ypresian (middle early Eocene) age for the Baja Wasatchian mammal assemblage. Although these results support previous correlations of North American Wasatchian faunas as early Eocene in age, they do not provide further resolution of the location of the Paleocene/Eocene boundary within the North American land mammal age sequence. An interesting result of the biochronologic correlations is that the Lomas las Tetas de Cabra ("Punta Prieta") Wasatchian land-mammal fauna probably is *not* earliest Wasatchian in age, even though a "stage of evolution" concept, on the basis of the presence of pantodonts and several relatively primitive species of characteristic Wasatchian genera, might have suggested an older age for the fauna. This study complements a series of recent studies integrating mammalian and marine biochronologies, magnetic polarity stratigraphy, and radioisotopic dating (Prothero and others, 1982; Berggren and others, 1985; Prothero and Armentrout, 1985; Flynn 1986). These integrated projects have rapidly increased our understanding of the temporal correlation and calibration of Paleogene marine and continental biochronologies.

The presence of diagnostic Wasatchian mammalian assemblages from Baja to the western interior (New Mexico, Colorado, Wyoming, Utah) to Ellesmere Island (Arctic Canada) documents the applicability of North American Land Mammal Ages across broad geographic areas. We are now able to support the hypothesis of the founders of the concept of land mammal ages (Wood and others, 1941) that these ages would be biochronologic units recognizable across all of North America and representing all of Tertiary time. Temporal correlations using mammalian faunal assemblages require synchrony of the biochronologic units (Flynn and others, 1984). Our results support synchrony of widely geographically separated Wasatchian Land Mammal Age assemblages (within the resolution of the available chronologic information).

The presence of a diagnostic restricted "Capay" Stage molluscan assemblage indicates that West Coast molluscan assemblages also are recognizable over wide geographic areas. In Baja California, as in other regions of the North American west coast, a "Capay" Stage molluscan assemblage is associated with a probable Penutian to Ulatisian benthic foraminiferal assemblage. This pattern of co-occurrence indicates synchrony of the molluscan assemblages, within the resolution of the benthic foraminiferal assemblages, across a broader geographic area than previously had been recognized. Early Cenozoic, West Coast molluscan stages are applicable at least from the Gulf of Alaska to central Baja California, a distance over which several faunal "provinces" are found today (controlled by contemporary water-mass boundaries). The broad geographic extent of early Eocene molluscan assemblages supports previous hypotheses of a warmer, more equable global climate.

The early Eocene paleolatitude of the Baja California peninsula was almost the same as its present latitude. These results conflict with previous studies, which suggested that the peninsula underwent northward translation during late Eocene–Oligocene, Miocene, or post-Miocene time. The results of previous studies, however, also conflicted with each other, yielding ambiguous and controversial conclusions as to the timing of motion of Baja. Some of these conflicts may be due to small data sets, incomplete averaging of secular variation, secondary magnetizations, and other magnetic remanence problems. The extensive study of mainland Baja Miocene volcanics by Hagstrum and others (1987) convincingly documented that Baja was in its pre–Gulf of California position since at least the early Miocene, contrary to the 10°-farther-south position suggested by Morris and others (1986). A more interesting, and more likely, cause of some of the discrepancies between these studies, however, may be the independent motion of smaller "terranes" that previously had been considered parts of single, larger, coherent terranes. For instance, some of the paleomagnetic results used to suggest the more southerly Miocene position of Baja by Morris and others (1986) are derived from the Transverse Ranges (Kamerling and Luyendyk, 1979; Hornafius, 1985) and the Channel Islands (Luyendyk and others, 1985). Both the Transverse Ranges and the Channel Islands lie on different terrane blocks (of Blake and others, 1982) than does the mainland Baja peninsula. Further, most of the Cretaceous paleomagnetic results from the variably termed "Baja-Borderland Allochthon" (BBA) or "Peninsular Ranges Terrane" (PRT) were derived from the mainland Baja California peninsula (principally from the "Santa Ana Terrane" of Blake and others, 1982). Most subsequent paleomagnetic analyses of Cretaceous and Cenozoic rocks assigned to the BBA/PRT were from islands offshore from California, again in terranes different than the mainland of Baja. Most importantly, the only other study determining the early Cenozoic motion of the

BBA/PRT is from San Miguel Island (Champion and others, 1986). San Miguel Island lies several terranes seaward of the mainland Baja "Santa Ana/Cortez" terrane, at the northernmost edge of the "Nicolas terrane" (very close to the suture with the Santa Lucia–Orocopia Allochthon). Discordant paleomagnetic directions within the "Baja-Borderland Allochthon/Peninsular Ranges Terrane" probably reflect the lack of cohesion of these major allochthons, and independent movement of the included smaller terranes throughout the Cenozoic. It is obvious that we must treat the smaller terranes as discrete, independent regions. Careful analysis of the paleomagnetic directions from the independent terranes of southern California, Baja California, and the eastern Pacific should give a better picture of the motion of these plate-margin blocks.

Our paleomagnetic results provide the most reliable early Cenozoic constraint on motion and paleolatitude of the Baja peninsula. Clearly, the mainland Baja California peninsula had reached its present latitude, and pre–Gulf of California position adjacent to Mexico, prior to the early Eocene. Further constraint on the timing of northward translation of the peninsula will be provided only by detailed studies of latest Cretaceous and Paleocene rocks.

CONCLUSIONS

(1) The mammalian fauna from the Lomas las Tetas de Cabra area, Rosarito quadrangle, ("Punta Prieta" fauna) clearly is Wasatchian in age, rather than Clarkforkian. This Wasatchian assemblage is remarkably similar to other Wasatchian faunas from the United States western interior, at the generic level.

(2) Extensive molluscan assemblages from marine strata intertonguing with terrestrial strata in the Santa Rosalia quadrangle can be correlated to the restricted "Capay" Stage (of Givens, 1974). This assignment is based on the presence of three "Capay" Stage index taxa, six taxa with "Capay" Stage first occurrences, and six taxa with "Capay" Stage last appearances. The "Capay" Stage has been correlated previously with planktonic foraminiferal zone early P7 to late P8 and calcareous nannoplankton zone mid-CP9b to early CP11 and therefore is middle Ypresian (middle early Eocene) in age.

(3) Benthic foraminifera from these marine strata can best be assigned to an undifferentiated Penutian/Ulatisian stage; this correlation supports a middle Ypresian age assignment for these sections. Further support is provided by the presence of the ostracode *Buntonia*, which has an early Eocene first appearance.

(4) All of the paleomagnetic sites from three sections are reversely magnetized. Correlation of this single reversed polarity magnetozone to a standard magnetic polarity time scale cannot be achieved on the basis of pattern alone. Available biochronologic information from these sections, however, indicates a middle Ypresian age and a likely correlation of this reversed polarity interval to Chron C23R or the reversed polarity zone within Chron C24N.

(5) This study provides the first direct integration of earliest Paleogene marine and terrestrial biochronologies in North America. The Wasatchian North American Land Mammal Age is at least partially correlative with the Ypresian Age, as had been suggested previously from less direct evidence.

(6) Wasatchian mammalian assemblages now are known from Ellesmere Island, Canada (78°N), the United States western interior (Wyoming to New Mexico, 35°–45°N), southern California (32°N), and Baja California, Mexico (28.5°N). All available geochronologic information indicates that Wasatchian mammalian assemblages are synchronous across broad, intracontinental geographic areas. Previous models of geographic faunal asynchrony, and earlier origination of Wasatchian mammalian assemblages either in the Arctic or Central America, are contradicted by our results.

(7) Our early Eocene paleomagnetic study provides the first early Cenozoic paleomagnetic results from the Baja peninsula. Plate-tectonic motion of Baja California during the early Cenozoic previously had been poorly constrained. Paleolatitudes calculated from our results indicate that Baja California was in its pre–Gulf of California position, sutured adjacent to the northern Mexican mainland, prior to the early Eocene. Previous studies had suggested that Baja underwent post-Cretaceous northward translation of ~10°, either (1) post–late Miocene (Morris and others, 1986), or (2) pre-Miocene, but probably late Eocene to Oligocene (Hagstrum and others, 1985, 1987; Champion and others, 1986). Conflicts between these studies may be due partly to paleomagnetic remanence problems. Many of these "Baja-Borderlands Allochthon/Peninsular Ranges Terrane" paleomagnetic results, however, are derived from terranes other than the mainland Baja "Santa Ana/Cortez" terrane. A more likely explanation of many of the discrepancies is the failure to treat small terranes as discrete, independent, mobile land masses.

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